secure communications between services, and between users and applications. OAuth 2.1 and OpenID Connect will be the standards you'll rely on to implement such functionality. Spring Security supports those standards and integrates seamlessly with external services to provide authentication and authorization. You'll use Keycloak for identity and access control management.

Also, I'll introduce the concepts of secrets management and encryption. I won't be able to go too deep into those topics, but I'll show you how to manage secrets to configure Spring Boot applications.

TESTING

Automated testing is paramount for the success of a cloud native application. A few levels of automated tests will cover the Polar Bookshop application. I'll show you how to use JUnit5 to write unit tests. Spring Boot adds many convenient utilities that improve integration testing, and you will use them to ensure the quality of your services. You'll write tests for the various features used in Polar Bookshop, including REST endpoints, messaging streams, data integrations, and security.

Keeping parity across environments is essential to ensure the quality of your applications. This is especially true when it comes to backing services. In production, you will use services such as PostgreSQL and Redis. During testing, you should use similar services rather than mocks or test-specific tools like the H2 in-memory database. The Testcontainers framework will help you use real services as containers in your automated tests.

BUILD AND DEPLOYMENT